

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-Blind, Placebo-Controlled Study to Investigate the Efficacy of Fenebrutinib in Relapsing Multiple Sclerosis **Short Title/Acronym:** FENopta Study **Protocol Number:** GN43271 (also 2021-003772-14) **NCT Number:** NCT05119569 **Sponsor Name:** Hoffmann-La Roche (Genentech) **Investigational Product:** Fenebrutinib (R07010939) **Phase of Development:** Phase II **Medical Monitor:** Hoffmann-La Roche Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of fenebrutinib compared to placebo on the total number of new gadolinium (Gd)-enhancing T1 brain lesions observed on MRI scans of the brain over 12 weeks. **Secondary Objectives:** To evaluate the number of new or enlarging T2-weighted lesions, the proportion of participants free from any new Gd-enhancing T1 lesions and new or enlarging T2-weighted lesions, as well as to assess the safety and pharmacokinetics of fenebrutinib.

2.2 Background & Rationale To address the need for therapies that can suppress disease activity and prevent disability progression in relapsing forms of multiple sclerosis (RMS). Bruton’s tyrosine kinase (BTK) is essential for the inflammatory activity of B-cells and microglia, which contribute to MS pathology. Fenebrutinib is a highly selective, non-covalent, and reversible BTK inhibitor. By inhibiting both B cells and microglia, fenebrutinib has the potential to profoundly reduce MS disease activity and impact long-term disease progression.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: 109 adult participants **Number of Sites:** Global multicenter trial **Estimated Study Start:** March 2022 **Estimated Primary Completion:** September 2024 (Primary Double-Blind Phase completed earlier)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 55 years. 2. Diagnosis of Relapsing Multiple Sclerosis (RMS) in accordance with the revised 2017 McDonald Criteria. 3. Expanded Disability Status Scale (EDSS) score of 0 to 5.5 at screening. 4. Evidence of disease activity: At least 2 relapses in the last 2 years, OR 1 relapse in the last year, OR at least 1 active lesion visible on an MRI scan within the past 6 months.

4.2 Exclusion Criteria 1. Diagnosis of Primary Progressive Multiple Sclerosis (PPMS) or non-active Secondary Progressive Multiple Sclerosis (SPMS). 2. Any known or suspected active infection at screening, including positive screening tests for Hepatitis B and C, tuberculosis (TB), HIV, or progressive multifocal leukoencephalopathy (PML). 3. Pregnant or breastfeeding women, or intending to become pregnant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Fenebrutinib 200 mg (or matched placebo) administered as two 100 mg tablets twice a day. **Route of Administration:** Oral **Duration of Treatment:** 12 weeks for the double-blind treatment (DBT) phase, followed by an optional open-label extension (OLE) phase where all participants receive fenebrutinib for up to 192 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Routine symptom management therapies that do not interfere with the investigational product. Inhaled and topical corticosteroids are permitted. **Prohibited:** Adrenocorticotrophic hormone or systemic corticosteroid therapy within 4 weeks prior to screening, immunosuppressants, high-dose biotin, and live-attenuated vaccines within 6 weeks prior to randomization.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, EDSS assessment, Baseline MRI
Baseline	Day 0	Randomization, First Dose
Interim	Weeks 4, 8	Onsite Visits: Safety Labs (including strict liver function test monitoring), Efficacy Assessment, MRI scans
End of DBT Study	Week 12	MRI scan, clinical evaluation, AE assessment, transition to Open-Label Extension (OLE)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Total number of new gadolinium (Gd)-enhancing T1 brain lesions observed on MRI scans of the

brain over 12 weeks (calculated as the sum of new lesions observed at Weeks 4, 8, and 12). **Measurement Method:** Centralized reading center radiologic evaluation using a standardized MRI protocol.

7.2 Secondary & Exploratory Endpoints * Total number of new or enlarging T2-weighted lesions observed on brain MRI over 12 weeks. * Proportion of participants free from any new Gd-enhancing T1 lesions and new or enlarging T2-weighted lesions. * Pharmacokinetics of fenebrutinib and evaluation of adverse events (AEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs and SAEs. Specific focus on liver function tests (ALT, AST, total and direct bilirubin) and monitoring for suicidal ideation/behavior using the Columbia-Suicide Severity Rating Scale (C-SSRS).

Serious Adverse Events (SAEs): Standard reporting timelines. Protocol specifies interruption of study treatment for suspected severe events like progressive multifocal leukoencephalopathy (PML) or elevated liver enzymes. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (iDMC) established to review safety data, including dedicated safety data reviews during the trial to assess signals like hepatotoxicity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All randomized patients for efficacy endpoints (analyzed according to treatment assigned); Safety evaluable population for all safety analyses. **Sample Size**

Justification: Approximately 109 participants, adequately powered to detect a significant difference in the total number of new T1 Gd+ lesions relative to placebo over 12 weeks. **Handling of**

Missing Data: A hypothetical strategy estimand is utilized to estimate the treatment effect had the patient not experienced a protocol-defined intercurrent event. If models fail due to a high number of zero lesion counts, logistic regression models adjusting for stratification factors are utilized.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard clinical site monitoring with additional ad-hoc visits implemented for rigorous liver function test tracking. **Audit Trail:** eCRF completion, centralized reading for MRI scans to ensure unbiased efficacy assessments, and strict adherence to the independent Data Monitoring Committee charter.

11. Study Identifiers & Governance

Primary ID: GN43271 **Registry Number:** NCT05119569 **Sponsor/Lead Organization:** Hoffmann-La Roche (Genentech) **Study Title:** FENopta **Official Regulatory Title:** A Randomized, Double-Blind, Placebo-Controlled Study to Investigate the Efficacy of Fenebrutinib in Relapsing Multiple Sclerosis

12. Clinical Framework (MS Specific)

Primary Condition: Relapsing Multiple Sclerosis (RMS) **Diagnosis Threshold:** Revised 2017 McDonald Criteria and EDSS ≤ 5.5 **Medical Methodology:** Non-covalent, reversible Bruton's Tyrosine Kinase (BTK) inhibition **Optimization Target:** Suppression of acute inflammatory MRI lesions (Gd-enhancing T1 and enlarging T2) and clinical relapse prevention.

13. Study Procedures & Methodology

Management Pathway: Centralized MRI monitoring and strict clinical assessment scales (EDSS) **Education Protocol (HCP):** Site-specific training on standardized MRI acquisition and safety monitoring protocols (especially hepatic safety) **Education Protocol (Patient):** Counseling on the double-blind nature of the trial, medication adherence, and the optional Open-Label Extension (OLE) **Quality Audit Mechanism:** iDMC safety data reviews, centralized independent radiological review.

14. Observation & Follow-Up Schedule

Total Duration: 12 weeks (Double-Blind Phase) + up to 192 weeks (Open-Label Extension) **Onsite Visit Cadence:** Every 2 to 4 weeks during the initial 12-week DBT phase. **Remote Monitoring Frequency:** Continuous safety reporting via site contact **Checklist Review Interval:** Regular tracking of liver function, C-SSRS, and concomitant medications

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				